

No $\log n$ when the candidate exponents differ: the $d = 2$, $h = (0, 1)$ chart integral in closed form

Claude, with Daniel Murfet

2026-09-06

Abstract

The companion of the first-log n theorem. For $k = (1, 1)$, $h = (0, 1)$ the candidate exponents are $\mu_1 = \frac{1}{2} < \mu_2 = 1$, so the paper predicts leading exponent $\min = \frac{1}{2}$ with multiplicity 1 and no logarithm. We prove the *exact* chart formula

$$Z(n) = \int_0^b \int_0^b v e^{-\beta n(uv)^2 + \beta \sqrt{n} uv a} dv du = b n^{-1/2} F_0(L) - n^{-1} \frac{F_1(L)}{b}, \quad L = \sqrt{n} b^2,$$

with $F_0 = \int_0^x f$, $F_1 = \int_0^x s f(s) ds$, hence $|Z(n) - b A n^{-1/2}| \leq (2M/b) n^{-1}$ and $Z(n) \sim \frac{b}{2} S_{1/2}(a) n^{-1/2}$. Zero ~~sorry~~/axiom.

1 Setting

A second GPT-6 Astra consult (`tide-log/gpt6_gen2d_v1.md`) gave the general two-dimensional reduction $Z(n) = \frac{n^{-p_1/2} B_2^{p_2-p_1}}{k_1 k_2} \int_0^L x^{p_1-p_2-1} F_{p_2-1}(x) dx$ with $p_i = (h_i + 1)/k_i$, and the dichotomy: equal p_i gives $\int F/x \sim A \log L$ (unit 35), unequal p_i gives the exact noncritical identity $\int_0^L x^{p-q-1} F_{q-1} = \frac{L^{p-q} F_{q-1}(L) - F_{p-1}(L)}{p-q}$ and no logarithm. The instance $k = (1, 1)$, $h = (0, 1)$ ($p_1 = 1$, $p_2 = 2$) needs only the unweighted primitives F_0, F_1 , so it is the natural first noncritical case.

2 The things formalised

```
theorem quadPowerPrimitive_eq ( a L : ) (h : 0 < ) (hL : 0 < L) :
  quadPowerPrimitive a L = quadPrimitive a L - quadMomentPrimitive a L / L
theorem twoDIntegralH01_exact ... :
  twoDIntegralH01 a b n = b * (sqrt n) ^ 1 * F (sqrt n b^2) - n ^ 1 * b ^ 1 * F (sqrt n b^2)
theorem twoDIntegralH01_sub_le ... :
  |twoDIntegralH01 a b n - b * quadMass a * (sqrt n) ^ 1| ^ 2 * quadMoment a / b * n ^ 1
theorem twoDIntegralH01_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n => twoDIntegralH01 a b n) ~[atTop]
  fun n => b * fluctuation (1/2) a / 2 * n ^ (-(1/2 : ))
```

3 Proof strategy

The inner v -integral with weight v is $c^{-2} F_1(cb)$, $c = \sqrt{n} u$ (`integral_comp_mul_left` with the kernel $x \mapsto x f(x)$); the outer integral becomes $(b/\sqrt{n}) K(\sqrt{n} b^2)$ with $K(L) = \int_0^L x^{-2} F_1(x) dx$.

The identity $K(L) = F_0(L) - F_1(L)/L$ is proved without integration by parts at the singular endpoint: the difference G has derivative $x^{-2}F_1 - f + (x \cdot xf - F_1)/x^2 = 0$ on $(0, \infty)$ (the primitive derivatives from `Continuous.integral_hasStrictDerivAt`, that of K from `intervalIntegral.integral_hasDerivAt_right` since the integrand is bounded near 0 and continuous at $L > 0$), so $G(t) = G(1)$ by `integral_eq_sub_of_hasDerivAt` on $[1, t]$; and $|G(t)| \leq 2Et$ on $(0, 1]$ forces $G(1) = 0$. The asymptotic follows from $0 \leq A - F_0(L) \leq M/L$ (unit 35) and $F_1 \leq M$.

4 Roadblocks and resolutions

- `rw [← hss]` with `hss :  $\sqrt{n} * \sqrt{n} = n$` rewrote *every* n , including the one inside $\sqrt{n}$, producing $\sqrt{\sqrt{n}^2}$; instead normalise with `ring_nf` and rewrite `Real.sq_sqrt` forwards.
- `convert` on a `HasDerivAt` between a λ -function and a Pi-expression left an instance-mismatch goal; define G in Pi form $((K - F) + F / \text{id})$ and use `HasDerivAt.congr_deriv`, keeping a separate `hGapp` for the pointwise unfolding.
- The `IntervalIntegrable.mono_fun` bound arrives as an un-beta-reduced ; use `(ae_restrict_iff' measurableSet_uIoc).2, Set.uIoc_of_le`, then `beta_reduce`.
- `ContinuousAt.inv` needs `(fun x => id x ^ 2) x 0`; `positivity` cannot see through the `redex` — pass `pow_ne_zero 2 hx0.ne'` (`defeq`).
- After `abs_mul` splits a product of three factors, rewrite each factor’s absolute value with `simp` only rather than a positional `rw`.

5 What was Mathlib, what was new

Mathlib: `integral_hasDerivAt_right`, `Continuous.integral_hasStrictDerivAt`,
`integral_eq_sub_of_hasDerivAt`, `HasDerivAt.div`, `integral_id`,
`norm_integral_le_of_norm_le_const`, `integral_comp_mul_left`. New: the noncritical
power-primitive identity and the exact closed form of a two-dimensional standard integral in terms
of the kernel primitives, exhibiting multiplicity one.

6 Lessons

For identities between primitives, “zero derivative on $(0, \infty)$ plus a bound at 0^+ ” is a robust substitute for integration by parts with a singular endpoint. Together with unit 35 this gives both halves of the two-dimensional multiplicity criterion in concrete instances.

7 Follow-ups

General (k_1, k_2) , (h_1, h_2) via the weighted primitive F_γ (real weights, power substitutions), the next-order term in the log case, and $d \geq 3$.